ARM Operating Modes — Exam Ready Notes

Source: Module 5 “The ARM Instruction Set” p.13 (plus standard CPSR facts). Highlighted italic lines are **comments**.

1. The seven processor modes (module verbatim)

Mode	Entered when	Class
User	Unprivileged mode under which **most tasks** run (normal application code).	Unprivileged; no SPSR
FIQ	A **high-priority (fast) interrupt** is raised.	Privileged exception mode; own SPSR
IRQ	A **low-priority (normal) interrupt** is raised.	Privileged exception mode; own SPSR
SVC	On **reset** and when a **Software Interrupt (SWI)** instruction is executed.	Privileged exception mode; own SPSR
Abort	A **memory access violation** occurs.	Privileged exception mode; own SPSR
Undef	An **undefined instruction** is fetched.	Privileged exception mode; own SPSR
System	Entered deliberately (OS tasks); **privileged mode using the same registers as User mode**.	Privileged; no SPSR

*Comment: remember the modes as **1 user + 5 exception + 1 system**. Only User is unprivileged; the five exception modes each get a private SPSR; System is “User’s registers with privilege”.*

2. Grouping the modes (classic exam view)

- **Privileged vs unprivileged:** all modes except User are privileged — they may use the MSR instruction to change mode and can access all system resources and the banked registers.
- **Exception vs non-exception modes:** FIQ, IRQ, SVC, Abort and Undef are **exception modes** — hardware switches into them automatically and each has a **banked SPSR** to save the previous CPSR. User and System are non-exception modes (no SPSR).
- **How the mode changes:** hardware on reset / interrupt / fault; or software via an **MSR** write to the mode bits of CPSR (privileged modes only).

Comment: hardware mode switching + banked registers is why ARM exceptions are fast: the new mode gets its own sp/lr/spsr, so the old context is preserved without saving anything first.

3. CPSR mode bits (standard fact)

Mode	M4	M3	M2	M1	M0	Hex (M[4:0])
User	1	0	0	0	0	0x10
FIQ	1	0	0	0	1	0x11
IRQ	1	0	0	1	0	0x12
SVC	1	0	0	1	1	0x13
Abort	1	0	1	1	1	0x17
Undef	1	1	0	1	1	0x1B
System	1	1	1	1	1	0x1F

The CPSR also carries the condition flags **N, Z, C, V**, the interrupt disables **I** (IRQ) and **F** (FIQ) and the state bit **T** (ARM/Thumb).

Comment: “which bits of CPSR select the mode?” is a 2-mark favourite — answer M[4:0], and quote two encodings (User 10000, SVC 10011) as proof.

4. Banked registers per mode (link to register sets)

- FIQ banks **r8–r14** → its handler barely saves anything, which is why FIQ is the **fast** interrupt.
- IRQ, SVC, Abort and Undef each bank only **r13 (sp)** and **r14 (lr)**, plus a private **SPSR**.
- User and System share one register set; **r0–r12, r15 and CPSR** are common to every mode.

Comment: in an exam answer, pair the mode table with this banking table — they are two halves of the same question.

5. One-minute revision

Numbers: 7 modes · 6 privileged · 5 exception modes (5 SPSRs) · 1 unprivileged (User) · mode = CPSR M[4:0].

Likely questions: (1) List the ARM operating modes and when each is entered. (2) Distinguish privileged/unprivileged and exception/non-exception modes. (3) Explain how the processor changes mode. (4) Why are registers banked per mode?

Comment: for full marks name the entry event of every mode — reset/SWI → SVC is the one most students forget.